PANJAB UNIVERSITY, CHANDIGARH

(Estted. under the Panjab University Act VII of 1947- enacted by the Govt. of India)

FACULTY OF ARTS

SYLLABI

FOR

MASTERS IN REMOTE SENSING AND GEOGRAPHIC INFORMATION SYSTEMS (GIS) (SEMESTER SYSTEM)

EXAMINATIONS, 2013-2014

PANJAB UNIVERSITY, CHANDIGARH

OUTLINES OF TESTS, SYLLABI AND COURSES OF READING FOR MASTERS IN REMOTE SENSING AND GEOGRAPHIC INFORMATION SYSTEMS (GIS) (SEMESTER SYSTEM) FOR THE EXAMINATIONS OF 2013 -14.

COURSE STRUCTURE

SEMESTER I **Max. Marks**

Paper I: Fundamentals of Remote Sensing, Photogrammetry & GPS **200**
- (a) Theory : 100
- (b) Practicals : 60
  (Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)
- (c) Internal Assessment : 40

Paper II: Fundamentals of Geographic Information Systems (GIS) **200**
- (a) Theory : 100
- (b) Practicals : 60
  (Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)
- (c) Internal Assessment : 40

Paper III: Image Processing **200**
- (a) Theory : 100
- (b) Practicals : 60
  (Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)
- (c) Internal Assessment : 40

SEMESTER II

Paper IV: Remote Sensing and GIS in Human Geography (with special reference to Urban and Regional Planning) **200**
- (a) Theory : 100
- (b) Practicals : 60
  (Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)
- (c) Internal Assessment : 40

Paper V: Remote Sensing and GIS in Physical Geography (with special reference to Environmental Management) **200**
- (a) Theory : 100
- (b) Practicals : 60
  (Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)
- (c) Internal Assessment : 40

Paper VI: Cartography **100**
- (a) Theory : 50
- (b) Practicals : 30
  (Lab. work 15 marks+ Record file 10 marks+ Viva 5 marks)
- (c) Internal Assessment : 20

Field Report **100**
- (a) Evaluation of the Field Report : 50
- (b) Presentation : 30
- (c) Viva Voce : 20

SEMESTER III

Paper VII:	Application of Remote Sensing and GIS in Disaster Management	**200**
	(a) Theory : 100	
	(b) Practicals : 60	
	(Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)	
	(c) Internal Assessment : 40	
Paper VIII:	Spatial Analysis	**200**
	(a) Theory : 100	
	(b) Practicals : 60	
	(Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)	
	(c) Internal Assessment : 40	
Paper IX:	Research Methodology and Report Writing	**200**
	(a) Theory : 100	
	(b) Practicals : 60	
	(Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)	
	(c) Internal Assessment : 40	

SEMESTER –IV

	Project Report	**600**
	(a) Mid-Term Appraisal of the Project Report : 100	
	(b) Evaluation of the Project Report : 400	
	(c) Presentation &Viva Voce (50+50) : 100	
	Grand Total	**2400**

NOTE:

1. Those candidates, who wish to discontinue after successful completion of first two semesters of the Masters course shall be awarded Diploma in Remote Sensing and Geographic Information Systems (GIS).
2. The minimum marks required to pass the examination shall be 45% in each (i) theory paper, (ii) practical work, and (iii) field/project report, and 50% in aggregate.
3. The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department, and one additional faculty member teaching the course.

SEMESTER - I

PAPER - I: FUNDAMENTALS OF REMOTE SENSING, PHOTOGRAMMETRY & GPS

Marks: 200

(a) THEORY **Marks: 100**
Time: 3 Hours

OBJECTIVES:

- to introduce the students to the basic principles of remote sensing;
- to indicate the methods of visual and digital interpretation of aerial photographs and satellite imageries;
- to explain the methods of acquiring remote sensing data and their usage; and
- to train the students about elementary photogrammetry and use of GPS.

COURSE CONTENTS:

UNIT – I

Introduction to remote sensing: Energy and radiation principles, energy-atmosphere interaction, energy-earth surface features interaction, spectral signatures.
Energy Recording Technology: Aerial Photography; Electronic Imaging Devices: Matrix Array, Pushbroom, Whiskbroom. Remote Sensing Detectors
Resolution of Remote Sensing data: Spatial, Spectral, Radiometric andTemporal.

UNIT – II

Type of aerial photographs and satellite imageries: Classification, characteristics and applications.
Sensor Platforms: Ground based, air borne and space borne, earth resources satellite systems: LANDSAT, SPOT, IRS, ERS, IKONOS, QUICKBIRD.

UNIT – III

Photogrammetry: Simple geometry of vertical photograph measurement of scale, relief displacement, parallax and height from vertical photographs, Flight Planning.

UNIT – IV

GPS-System Description, Historical Background, GPS Segment, Features of GPS Satellite, Principles of Operation, Present Status.
Surveying with GPS-Methods of observation, Absolute Positioning, Relative Positioning, Differential GPS, Kinematic GPS.
GPS Receivers: Navigation Receivers, Surveying and Mapping Receivers, Geodetic Receivers; Computation of coordinates; Application of GPS

NOTE:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit, and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

(b) PRACTICAL **Marks: 60**

1. Arrangement of aerial photographs with the help of numbers given on the photograph and preparation of photo index.
2. Interpretation of information given on the margins of the photographs and satellite imageries.
3. Stereo vision test, selection of stereopair, and its orientation and placing under the stereoscope for stereovision.
4. Identification and mapping of features from a single photograph
5. Identification and mapping of features from a stereopair and their verification in the field.
6. Locating principal point, conjugate principal point and flight line on a stereopair.
7. Measurement of scale, relief displacement, height and slope from vertical photographs.
8. Removing error due to relief displacement on vertical photographs by graphical method.
9. A Field based survey on GPS.

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

(c) INTERNAL ASSESSMENT **Marks: 40**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the university.

BOOKS RECOMMENDED

ESSENTIAL READINGS:

American Society : *Manual of Remote Sensing*, Vol. 1 and II, Falls Church of Photogrammetry, Virginia, 1975.

Barret, E.C. & Curtis, E.F. : *Remote Sensing of Environment*, Second Edition, 1982.

Curran, Paul J. : *Principles of Remote Sensing*, Longman, Hongkong, 1988.

Jensen, J.R. : *Introductory Digital Image Processing: A Remote Sensing Perspective*, Prentice Hall, New Jersey, 1986.

Lillesand, T.M. & Kiefer, : *Remote Sensing and Image Interpretation*, Third R.W. Edition, John Wiley and Sons, 1994.

FURTHER READINGS:

Agarwal, C.S. and P.K. Garg : *Text Book on Remote Sensing in Natural Resources Monitoring and Management*, Wheeler Publishing Co, New Delhi, 2000.

Jain, A.K. : *Fundamentals of Digital Image Processing*, Prentice Hall, New Jersey, 1989.

Muralikrishna, I.V. : *Remote Sensing Applications and Geographic Information Systems Recent Trends*, Tata McGraw Hill, New Delhi, 1992.

Fratt, W.K. : *Digital Image Processing*, John Wiley & Sons, 1978.

Rashid, S.K.	:	*Remote Sensing in Geography*, Manak Publications, Delhi, 1993.
Rees, W.G.	:	*Physical Principles of Remote Sensing*, Cambridge University Press, Cambridge, 1990.
Richards, John, A.	:	*Remote Sensing*: Digital Image Analysis, Springer Verlag, 1986.
Swain, Phillip, H.	:	*Remote Sensing*: The quantitative Approach, McGraw Hill International Book Co., 1978.
Verbyla, David, L	:	*Satellite Remote Sensing of Natural Resources*, Lewis Publishers, New York, 2005

PEDAGOGY:

- Students may be taken to field for verification of identifiable features on aerial photographs and satellite imageries;
- Students may be taken to different institutions, such as IIRS, NRSA, State Remote Sensing Centres to acquaint with different equipments, techniques, and their products; and
- Students may be asked to prepare a report using toposheets, aerial photographs and satellite imageries to trace the temporal and spatial changes.

PAPER II: FUNDAMENTALS OF GEOGRAPHIC INFORMATION SYSTEMS (GIS)

Marks: 200

(a) THEORY **Marks: 100**

Time: 3 Hours

OBJECTIVES:

The main objective of this course is to expose the students to fundamental principles of Geographical Information Systems including basic concepts and definitions, methods and techniques.

COURSE CONTENTS:

UNIT – I

Concepts and Definitions:

- Geographic Information Systems (GIS): Definition and developments
- Spatial data base: Points, Lines and Polygons
- Computer components: Hardware and software

UNIT – II

Functional Elements:

- Data acquisition, input, editing, data manipulation and topology creation
- Data analysis and query
- Symbolisation and product generation

UNIT – III

Data Management and Structure:

- Data Base Management System (DBMS): Purpose and structure
- Classification: Raster, Vector, quadtree, hierarchical, network and relational

UNIT - IV

Human and Organizational Issues:

- GIS Applications, GIS users, Hardware and Software questions for organization considering GIS.
- GIS: Decision Support System

NOTE:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit, and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

(b) PRACTICAL **Marks: 60**

1. Windows/ NT Environment: Basics, file and directory organisation and management.
2. Familiarization with GIS Software
3. Representation of Real World in GIS: Point, Line and Polygon.
4. Linking Spatial and Attribute Data
5. Spatial data analysis techniques: Buffering, Overlay and Querying.

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

(c) INTERNAL ASSESSMENT **Marks: 40**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the university.

BOOKS RECOMMENDED

ESSENTIAL READINGS

Aronoff, S. : *Geographic Information Systems: A Management Perspective*, WDL Publications Ottawa,Canada, 1992.

ESRI : *Understanding GIS*, Environmental Systems Research Institute, U.S.A., 1993.

Jefrey S. & John, E. : *Geographic Information Systems - An Introduction*, Prentice Hall, New Jersey, USA, 1990.

Michael F. Goodchild : *Introduction to GIS*, Santa Barbara, California, & Karan K. Kemp (eds.) NGGIA, 1990.

Ian Heywood,Sarah Cornelius, Steve Carver : An Introduction to Geographical Information Systems, Prentice Hall College Div, New Jersey, USA,2000.

Kang-tsung (Karl) Chang : Introduction to Geographic Information Systems, 4th Edition, Tata McGraw-Hill, New Delhi, 2009.

FURTHER READINGS

Elaine, Rich : *Artificial Intelligence*, McGraw Hill Book Co., 1983.

Giarratano, J. & Rieley, G. : *Expert Systems: Principles and Programming*, PRS Kent Publishing Co., 1985.

Laurini, R. & Thomson, D. : *Fundamentals of Spatial Information Systems*, Academic Press, London, 1994.

Rashid, S.M. : *Remote Sensing in Geography*, Manak Publications, Delhi, 1993.

PEDAGOGY:

Taking students for fieldwork, inviting professionals for lectures, taking students to institutes/training centres engaged in imparting professional training and providing research consultancy in business, tourism and area/regional planning should be the main focus.

Paper III: IMAGE PROCESSING **Marks: 200**

(a) THEORY **Marks:100**
Time: 3 Hours

OBJECTIVES:
The course is designed to introduce the student to analytical tools and methods which are currently used in digital image processing as applied to image information for human viewing. Then apply these tools in the laboratory in image restoration, enhancement and compression.

UNIT – I

Image processing: definition and types-visual and digital;
Visual image interpretation: Image characteristics, elements of object identification, image interpretation techniques, field verification.

UNIT - II

Digital Image Pre-processing: Sources of Errors, Types of error-Radiometric and Geometric, Image Rectification: Geometric Correction, Radiometric Correction, and Noise Removal.

UNIT - III

Image Enhancement Techniques: Contrast enhancement-Linear and Non-Linear, Logarithmic and Exponential, Gaussian stretch, Density slicing,
Spatial Filtering: Low frequency and High frequency, Edge enhancement, Band ratioing and band combination, Principal Component Analysis.

UNIT - IV

Digital image classification: Classification scheme-Supervised classification (training site selection and statistical information extraction),
Discriminate functions: Maximum Likelihood classifier, Euclidian distance; Unsupervised Classification, Classification accuracy assessment, Error matrix.

NOTE:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

(b) PRACTICAL **Marks: 60**

(i) Familiarization with image processing system
(ii) Loading of image data, identification of objects on video display, study of histograms
(iii) Image enhancement techniques: contrast enhancement, band ratioing, edge enhancement, filtering, density slicing
(iv) Image registration: image to map, image to image
(v) Image classification techniques: supervised and unsupervised

(vi) Accuracy analysis
(vii) Ground data collection for training sets for classification of image

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

(c) INTERNAL ASSESSMENT **Marks: 40**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the university.

BOOKS RECOMMENDED

ESSENTIAL READINGS

Burger W., & Burge, M.J. : *Principles of Digital Image Processing,* Springer-Verlag London Limited, 2009.

Gonzalez, R.C. & Woods, R.E. : *Digital Image Processing*, Second edition, Prentice-Hall, Inc., 2002.

Jensen, J.R. : *Introductory Digital Image Processing: A Remote Sensing Perspective*, Prentice Hall, New Jersey, 1986.

Mather, P.M. : *Computer Processing of Remotely-Sensed Images: An Introduction,* Third Edition, John Wiley & Sons, 2008.

FURTHER READINGS

Lillesand, T.M. & Kiefer, R.W. : *Remote Sensing and Image Interpretation*, Third Edition, John Wiley and Sons, 1994.

Pratt, W.K. : *Digital Image Processing*, Third Edition, John Wiley & Sons, 2001.

Prithvish Nag and M. Kudrat : *Digital Remote Sensing*, Concept Publishing Company, New Delhi, 1998.

Jain, A.K. : *Fundamentals of Digital Image Processing,* Prentice-Hall, Inc., 2002.

PEDAGOGY:

Students may be asked to prepare a report on image processing techniques using satellite images. Students will be encouraged to ask questions and participate in classroom discussions.

SEMESTER II

PAPER IV: REMOTE SENSING & GIS IN HUMAN GEOGRAPHY **Marks: 200**
(with special reference to Urban and Regional Planning)

(a) THEORY **Marks: 100**
Time: 3 Hours

OBJECTIVES:
To provide an understanding of:

- concepts and definitions of urban and regional planning so as to pin point the specific areas, where application of Remote Sensing and GIS technologies will increase efficiency and effectiveness of planning;
- data base creation in terms of various attributes, types and hierarchical scales; and
- planning techniques which help in the location of infrastructural facilities for decentralised urban and regional planning.

COURSE CONTENT:

UNIT – I

Concepts and definitions: Planning - definition and types; role of remote sensing and GIS in regional planning, tourism and business management.

UNIT – II

Regional Planning: Creation of data base for regional planning; land use/ land cover mapping; land use suitability evaluation; updating of land use/ land cover maps; resource analysis for district/ block planning; location of amenities and facilities.

UNIT – III

Creation of data base for urban planning; updating of urban land use maps; location of facilities and amenities, urban growth monitoring, and urban change detection; route location and site planning, urban development planning; slum identification and improvement.

UNIT – IV

Urban and regional mapping, Base map preparation: regional city and intra–city.
Urban and regional plans formulation: Application of remote sensing & GIS techniques, Case Studies.
Urban analysis: Case studies pertaining to urban growth monitoring, change detection and slum development.

NOTE:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).
2 A total of eight questions will be set out of the whole syllabus, at least 2 from each unit and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

(b) PRACTICAL **Marks: 60**

(i) Image/photo interpretation: recognition and detection of objects, preparation of land use maps.
(ii) Digital image processing techniques in preparation of land use and urban land use maps.
(iii) Preparation of base map and its updating.
(iv) Urban growth monitoring and change detection.
(v) Slum identification and mapping.

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

(c) INTERNAL ASSESSMENT **Marks: 40**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the university.

BOOKS RECOMMENDED

ESSENTIAL READINGS

Avery, T.V. : *Interpretation of Aerial Photography,* Burgess Publishing Co., Minneapolis, 1962.

Barret, E.C. & Curtis, E.F. : 'The Built Environment' in Remote Sensing of Environment, Second edition, pp. 293-306, 1982.

Gautam, N.C. : *Urban Landuse Study through Aerial Photo Interpretation Techniques*, Pink Publishing House, Mathura, 1970.

Muralikrishna, I.V. : *Remote Sensing Applications and Geographic Information Systems - Recent Trends*, Tata McGraw Hill, New Delhi, 1992.

Nag, Prithvish : Thematic Cartography and Remote Sensing, Concept, New Delhi, 1992.

Richardson, B.F(ed) : Introduction to Remote Sensing of the Environment, Kendall/Hunt, Dubuque, Iowa, 1978.

Ian Heywood,Sarah Cornelius, Steve Carver : An Introduction to Geographical Information Systems
Prentice Hall College Div, New Jersey, USA,2000.

Kang-tsung (Karl) Chang : Introduction to Geographic Information Systems, 4th Edition, Tat McGraw-Hill, New Delhi, 2009.

FURTHER READINGS

Sundaram, K.V. : Urban and Regional Planning in India, Concept, New Delhi, 1977.

Taylor, John,L. : Urban Planning Practice in Developing Countries, Williams, David C Pergamon Press, 1981.

PEDAGOGY:

Taking students for fieldwork, inviting professionals for lectures, taking students to institutes/training centres engaged in imparting professional training and providing research consultancy in business, tourism and area/regional planning should be the main focus.

PAPER V: REMOTE SENSING AND GIS IN PHYSICAL GEOGRAPHY **Marks: 200**
(with special reference to Environmental Management)

(a) THEORY **Marks: 100**
Time: 3 Hours

OBJECTIVES:

- to train the students to detect, identify and recognise the various facets of different landforms using remote sensing and creating a data base in GIS mode;
- to educate students about morphogenetic analysis using photogrammetry & GIS; and
- to indicate the use of remote sensing and GIS in assessment and management of environmental problems.

COURSE CONTENT:

UNIT - I

Concepts and definitions:

- Geomorphic Processes, Terrain Elements, Principles of Geomorphological image interpretation, Identification and Reognition of landform
- Qualitative and quantitative study of relief.

UNIT – II

Remote Sensing and GIS in Geomorphology:

- Manual and digital methods of morphometric analysis of a drainage basin.
- Use of remote sensing in understanding the relationships between structure and process of landforms, climate and landforms structure and drainage pattern, landforms and surface material, landforms and landuse, landforms and human activities.
- Classification of landforms using remote sensing and GIS

UNIT – III

Remote Sensing and GIS in climatology, hydrology and environmental geomorphology

- Use of remote sensing and GIS in recording the meteorological data and weather forecasting.
- Use of remote sensing and GIS in hydrological studies, especially mapping of water resources and water related problems.
- Use of Remote Sensing & GIS in impact assessment of human activities on the geomorphological processes and the landscape

UNIT – IV

- Use of Remote Sensing and GIS in assessment of environmental problems such as deforestation, water logging, desertification, water, pollution etc.

NOTE:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

(b) PRACTICAL **Marks: 60**

(i) Identification and mapping of relief, drainage pattern, water bodies and surface material from topographical maps
(ii) Identification and mapping of relief, drainage pattern, water bodies and surface material from aerial photographs and satellite imageries.
(iii) Identification and mapping of terrain units from large scale aerial photographs
(iv) Measurement and mapping of slopes from topographical maps
(v) Rough estimation and mapping of slopes from the aerial photographs
(vi) Morphometric analysis of drainage basins from topographical maps
(vii) Analysis of certain selective aspects of the morphometry of the drainage basins from the aerial photographs and satellite imageries
(viii) Empirical quantitative classification of terrain from topographical maps
(ix) Genetic and functional classification of terrain from the aerial photographs and satellite imageries
(x) Identification and mapping of impact of human activities on the geomorphological processes and landscape using remote sensing and GIS techniques

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

(c) INTERNAL ASSESSMENT **Marks: 40**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the university.

BOOKS RECOMMENDED

ESSENTIAL READINGS

Agarwal, C.S. and P.K. Garg : *Text Book on Remote Sensing in Natural Resources Monitoring and Management*, Wheeler Publishing Co, New Delhi, 2000.

Druary, S.A. : *Image Interpretation in Geology*, Allen and Unwin Ltd., London, 1987.

Siegal, B.S. & Gillespie, A.R : *Remote Sensing in Geology*, John Wiley Publications, 1986.

Townsend, J.R.G. : *Terrain Analysis and Remote Sensing*, George Allen and Unwin, 1981.

Verstappen, H. : *Remote Sensing in Geomorphology*, Elseiver Scientific Publications, Netherlands, 1977.

Verbyla, David, L : *Satellite Remote Sensing of Natural Resources*, Lewis Publishers, New York, 2005.

FURTHER READINGS

Greedry, Alan, F.	:	Application of Remote Sensing with Special Reference to Geosciences, Gregory Geo-science, 1974.
Nefedov, K.E.	:	Deciphering of Groundwater from Aerial Photographs, Amerind Publications, Delhi, 1972.
Papova, T.A. and Pandey, S.N.	:	Principles and Applications of Photogeology, Wiley Eastern Publishers, 1987.
Smith, Willam, L.	:	Remote Sensing Applications for Mineral Exploration, Dawden Hutchintons and Ross Inc.,1977.

PEDAGOGY:

Students may be taken to institutions such as IIRS, NRSA, State Remote Sensing Cetres to acquaint them with equipments, techniques and their products. Students may be asked to prepare a report on landform using topographical sheets, aerial photographs and satellite images. Students may acquaint with the satellite imageries of various kinds of environmental problems.

PAPER VI: CARTOGRAPHY **Marks: 100**

(a) THEORY **Marks: 50**
Time: 3 Hours

OBJECTIVES:

The course is designed to emphasise the relevance and scope of cartography in GIS. It shall highlight the value of this vital field in the current and highly versatile field of GIS.

UNIT – I

- Cartography: Definition, Scope and Historical Perspective.
- Cartography and GIS
- Traditional vs Computer (Digital) Cartography

UNIT – II

- Types of Data: Spatial data and Attribute data
  Spatial Data: Point, line, area
  Attribute Data: Data Analysis and Classification
- Map Types: Quantitative and Qualitative

UNIT – III

- Map Design: Principle of Symbolization , Colour and Colour Schemes, Map Lettering, Map Layout, Legend Presentation
- Topographic Maps: General Introduction, Types and Classification

UNIT – IV

- Techniques of Relief Representation and Interpretation: Understanding Hill Shading, Hachures and Contours
- Profiles (longitudinal and Transverse) and Slope Analysis

NOTE:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, atleast 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks (36 marks). These will be in addition to the compulsory question at serial number I.

(b) PRACTICAL **MARKS: 30**

Exercises:

1. Classification of topographical sheets on various scales as per Old Map Series of India
2. Classification of topographical sheets on various scales as per New Map Series of India

3. Interpretation of Physical and Cultural aspects of an area using SOI topographical sheet on 1:50,000 scale.
4. Creation of Digital Elevation Model (DEM) using topographical sheets.
5. Generation of Slope Map using DEM
6. Generation of Aspect Map using DEM
7. Generation Hill Shading Map using DEM
8. Profile generation using Topographical Sheets: Longitudinal Profile, Transverse Profile
9. Profile generation using DEM: Longitudinal Profile, Transverse Profile and Surface Profile
10. Generation of Thematic Maps: Dot Map, Choropleth Map, Proportionate Circles, Isopleth Map

Software Used: Arc GIS and ERDAS IMAGINE

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

(c) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the university.

BOOKS RECOMMENDED

ESSENTIAL READINGS

Cromley, R.G.	:	*Digital Cartography* ,Prentice- Hall of India, New Delhi, 1992.
Crampton, W.C.	:	*Mapping :A Critical Introduction to Cartography and GIS,* John Willy & Sons, New York, 2010.
Dent, B.D.	:	Cartography-*Thematic Map Design*, WBC McGraw–Hill, Boston, 1999.
John Krygier and Denis Wood	:	Making Maps: *A visual guide to map design for GIS,* Guilford Publications, New York, 2005.
Keates, J.S.	:	*Cartographic Design and Production*, Longman, London, 1998.
Misra,R.P.and Ramesh, A.	:	*Fundamental of Cartography*, Concept Publishing Company, New Delhi, 1989.
Monkhouse, F.J.	:	*Maps and Diagrams*, Methuen and Co., London, 1994.
Nag, P.	:	Thematic Cartography and Remote Sensing, Concept Publishing Co.,New Delhi ,1992
Rampal, K.K.	:	*Mapping and Compilation*, Concept Publishing Co., New Delhi ,1993.

Raisz, Erwin	:	*Principles of Cartography*, McGraw Hill, New York, 1962.
Robinson, A.H. and others	:	*Elements of Cartography*, John Willy & Sons, New York, 6th edition, 1992.
Singh, R.L.and Singh R.P.B.	:	*Elements of Practical Geography,* Kalyani Publishers , New Delhi, Reprint 2002

FURTHER READINGS

Birch, T.	:	*Maps- Topographical and Statistical*, Clarendon Press, Oxford, 1949.
Brown, L.A.	:	*The Story of Maps,* Cressit Press, London, 1951.
		Harley J.B. and David Woodward: *The History of Cartography* Volume 2, Book 2, The University of Chicago Press, 1995.
Ramamurthy, K.	:	*Maps Interpretation: India Landscapes Through Survey of India*, Topographic Maps, R.K. Mutt Road, Madras, 1982.
Wood Clifford H. and Keller C.	:	*Cartographic Design- Theoretical and Practical Perspectives*, John Wiley & Sons, 1996.

PEDAGOGY:

Students will be encouraged to ask questions and participate in classroom discussions. Assignments, term papers and group discussions will be the additional methods to make teaching more focused and interesting.

FIELD REPORT **Marks: 100**

Distribution of Marks:

a) Evaluation of the Field Report	:	50
b) Presentation	:	30
c) Viva Voce	:	20

NOTE:

The field report shall be prepared by the student on the basis of work done or experiences gained through visit(s) to Remote Sensing Centre(s)/Institute(s)/Field visit(s) in India. The student will submit three copies of his/her work ten days before the commencement of the theory examination of the said semester. Three examiners consisting of supervisor, Chairperson of the department and one more faculty member teaching the course shall evaluate the field report independently and the student shall be awarded the mean score of the three evaluations. The evaluation process shall have three-components:(i) evaluation of field report, (ii) presentation of data analysis and main finding before the board of examiners and (iii) Viva Voce.

SEMESTER III

PAPER VII: APPLICATION OF REMOTE SENSING AND GIS IN DISASTER MANAGEMENT **Marks: 200**

(a) THEORY **Marks: 100**
Time: 3 Hours

OBJECTIVES:
Main objectives of this course are to introduce the students with the nature, causes and effects of disasters and applications of GIS and Remote Sensing in mapping, mitigation and monitoring of hazards and disasters.

UNIT – I

i. Fundamental concepts of hazards and disasters
ii. Natural and human induced disasters: classification, types
iii. Historical perspective of disasters in India

UNIT – II

iv. Disaster management: concepts relating to the pre-disaster phase, emergency phase, post-disaster phase
v. Disaster management mechanism in India: agencies, plans and policies
vi. Role of remote sensing and GIS in disaster management

UNIT – III

vii. Remote sensing and GIS application in geological hazards: Landslides and Earthquakes.
viii. Remote sensing and GIS application in Hydro meteorological Hazards: Floods, Cyclones, Drought.

UNIT – IV

ix. Remote sensing and GIS application in Environmental hazards: Deforestation, Land Degradation and desertification, Forest fires
x. Case Studies: Earthquakes in India, Drought in India, Floods in Indo-Gangetic plains, Landslides in Himalayan region,

NOTE:
1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit, and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

(b) PRACTICAL **Marks: 60**

1. Major Disaster Events in India (1900-2004)
2. Earthquake Prone Districts in India
3. Cyclone Prone Districts in India
4. Flood Prone Districts in India
5. Northwest Himalayas: Landslides Analysis
6. Himachal Pradesh: Earthquake Analysis

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

(c) INTERNAL ASSESSMENT **Marks: 40**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the university.

BOOKS RECOMMENDED

ESSENTIAL READINGS

Greg Bankoff , Georg Frerks, Dorothea Hilhorst	:	*Mapping Vulnerability: Disasters, Development and People*, Earthscan Publications Ltd., 2004.
Monmonier mark,	:	*Cartographies of Danger: Mapping Hazards in America***,** University of Chicago Press, 1997.
Roy, P.S.; Van Westen, C.J.; Jha, V.K.; Lakhera, R.C. and Champati Ray, P.K.,	:	*Natural Disaster and their Mitigation: Remote Sensing and Geographical Information System Perspectives*, IIRS, DehraDun, Govt. of India, India, 2000.

FURTHER READINGS

Agarwal, C.S. and P.K. Garg	:	*Text Book on Remote Sensing in Natural Resources Monitoring and Management*, Wheeler Publishing Co, New Delhi, 2000.
Burrough, P.A. & Macdoneli, R.A.	:	*Principles of Geographic Information Systems,* Oxford University Press 2000.
Demers, M.N.	:	*Fundamentals of geographic Information Systems*, 2nd Edition, John Wiley and sons 2000.
Giarratano, J. & Rieley, G.	:	*Expert Systems: Principles and Programming*, PRS Kent Publishing Company 1985.
Muralikrishna, I.V.	:	*Remote Sensing Applications and Geographic Information Systems Recent Trends*, Tata McGraw Hill, New Delhi, 1992.
Rashid, S.K.	:	*Remote Sensing in Geography*, Manak Publications, Delhi, 1993.
Rees, W.G.	:	*Physical Principles of Remote Sensing*, Cambridge University Press, Cambridge, 1990.
Swain, Phillip, H.	:	*Remote Sensing*: The quantitative Approach, McGraw Hill International Book Co., 1978.
Verbyla, David, L	:	*Satellite Remote Sensing of Natural Resources*, Lewis Publishers New York, 2005.

PEDAGOGY:

Students may be asked to prepare a report on zonation of hazards and disasters using topographical sheets, aerial photographs and satellite images. Students may acquaint with the satellite imageries of various kinds of disasters.

PAPER VIII: SPATIAL ANALYSIS **Marks: 200**

(a) THEORY **Marks: 100**
Time: 3 Hours

OBJECTIVES:
The course is designed to introduce the student to spatial analysis capabilities of GIS. To understand the logical connections between attribute data and map features, and the operational procedures built on the spatial relationships among map features.

UNIT - I

Introduction to Spatial analysis:

- Significance of Spatial analysis.
- Using GIS for Spatial Analysis.
- Quantification of Spatial Analysis
- Single Layer operations
- Multiple layer operations

UNIT - II

Vector Based Spatial Data Analysis

- Overlay
  - Point-in-Polygon, Line-in-Polygon, Polygon-in-Polygon
- Logical operators and Conditional Operators
- Types of Spatial Operations in GIS
  - Identity, Intersect, Union
  - Clip, Erase, Reselect
- Buffering: Point, Line, Area buffers

Raster Based Spatial Data Analysis

- Local functions
- Focal functions
- Zonal Functions
- Global functions

UNIT - III

Point Pattern Analysis

- Descriptive Statistics of Point Features
- Spatial Arrangement
  - Nearest Neighbour Index
  - Quadrat Analysis,
- Spatial Autocorrelation

Network Analysis

- Network Data Model
- Path Determination
- Resource allocation/distribution analysis
- Utility Locating/ Siting analysis
- Shortest path

UNIT - IV

Spatial Modeling

- Model building
- Multiple Regression
- Logistic Regression
- Data Manipulation and Programming

Surface Analysis

- Digital Elevation Model (DEM) and Digital Terrain Model (DTM)
- Contouring, Hill Shading and Perspective View (3D viewing)
- Slope and Aspect Analysis

NOTE:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

(b) PRACTICAL **Marks: 60**

(i) Familiarization with Spatial Analysis Software.
(ii) Spatial analysis: Query, Identifying, measuring, query by location/attribute
(iii) Measuring Central tendency; Mean Centre, Weighted Mean Centre, Median Centre, Standard distance, Standard Deviation Ellipse
(iv) Quadrat Analysis, Nearest Neighbour statistics
(v) Spatial Autocorrelation of Point features; Geary's Ratio, Moran's I Index
(vi) Network analysis
(vii) Vector Analysis; Buffering: Point, Line, Area buffers
(viii) Overlay
(ix) Surface Analysis; Digital Elevation Model (DEM) and Digital Terrain Model (DTM), Contouring, Hill Shading and Perspective View (3D viewing), Slope and Aspect Analysis.

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

(c) INTERNAL ASSESSMENT **Marks: 40**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the university.

BOOKS RECOMMENDED

ESSENTIAL READINGS

Chang, Kang-tsung : *Introduction to Geographic Information Systems,* Sixth Edition, Tata McGraw-Hill, 2011.

Chou, Yue-Hong	:	*Exploring Spatial Analysis in Geographic Information Systems,* OnWord Press, 1997.
Laurini, R. & Thomson, D.	:	*Fundamentals of Spatial Information Systems*, Academic Press, 1994.
Wong, David W.S. & Jay Lee	:	Statistical Analysis of Geographic Information With ArcView GIS and ArcGIS, John Wiley and Sons, 2005.

FURTHER READINGS

Burrough, P.A. & Macdoneli, R.A.	:	*Principles of Geographic Information Systems,* Oxford University Press, 2000.
Demers, M.N.	:	*Fundamentals of geographic Information Systems*, 2nd Edition, John Wiley and sons, 2000.
Muralikrishna, I.V.	:	*Remote Sensing Applications and Geographic Information Systems Recent Trends*, Tata McGraw Hill, 1992.
Rashid, S.K.	:	Remote Sensing in Geography, Manak Publications, 1993.
Rees, W.G.	:	Physical Principles of Remote Sensing, Cambridge University Press, 1990.

PEDAGOGY:

Students may be asked to prepare a report on spatial analysis applications using external and self generated databases. Students will be encouraged to ask questions and participate in classroom discussions.

PAPER IX: RESEARCH METHODOLOGY AND REPORT WRITING

Marks: 200

(a) THEORY **Marks: 100**
Time: 3 Hours

OBJECTIVES:
The aim of this paper is to train the students about the various aspects of research writing so that they are in a position to write research reports independently.

COURSE CONTENT:

UNIT-I

- Meaning, objectives and Types of Research, Research approaches and significance of research.
- The research process: the broad problem area, preliminary data collection, problem, selection and definition, theoretical framework, hypothesis development and elements of research design.
- Experimental design: the laboratory experiment, variables, validity, types of experimental designs.

UNIT-II

- Data collection, Measurement, processing and analysis: collection, processing and analysis; measurement in research, operational definition,
- Measurement scales, scaling; scaling techniques, reliability and validity;
- Sources of data; data collection methods: interviewing, questionnaires, other methods of data collection; review of statistical data analysis.

UNIT-III

- Sampling: Need and purpose of sampling, population and sample, population frame, sampling with and without replacement, population parameters.
- Sampling theory–sampling distributions, parameter estimation, hypothesis testing. Sampling designs – probability and non-probability sampling

UNIT-IV

- Report writing: the research proposal, the report, integral parts of the report, steps involved in report writing,
- Types of reports, oral presentation, conclusions.

NOTE:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit; and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

(b) PRACTICAL **Marks: 60**

(i) Data generation from Topographical Maps, Aerial Photographs and Satellite Imageries and its analysis.

(ii) Data collection using a schedule and a questionnaire on a given problem.
(iii) Sampling exercises from the given spatial data.
(iv) Preparation of a questionnaire on a given problem.
(v) Exercises on Data Classification, Tabulation and Mapping.
(vi) Preparation of a report on a problem using RS & GIS.

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.
Practical record book should be submitted one week before the commencement of theory exams.

(c) INTERNAL ASSESSMENT **Marks: 40**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the university.

BOOKS RECOMMENDED

ESSENTIAL READINGS

Ackoff, Russell L.(1961) : The Design of Social Research, Chicago; University of Chicago Press.

Bailey, Kenneth D.(1978) : Methods of Social Research, New York

Berdie,Douglas R and Anderson, John F. : Questionnaires; Design and Use, Metuchen NJ. The Scarecrow Press, Inc.

Creswell, John W. (2008) : Research Design; Qualitative, Quantitative and Mixed Methods Approach, SAGE Publications, Los Angeles.

Kothari, C.R.(2008) : Research Methodology, Methods & Techniques New Age International Publisher, N. Delhi.

Kumar Ranjit (2010) : Research Methodology A step-by-step Guide for Beginners, SAGE Publications, Ltd. London (Third Edition).

Misra, H.N and V.P.Singh (1998) : Research Methodology in Geography, Social, Spatial and Policy Dimensions, Rawat Publications, N. Delhi.

Smith, H.W (1975) : Strategies of Social Research The Methodological Imagination, Prentice-Hall, Inc., Englewood Cliffs, New Jersey.

FURTHER READINGS

Bennet P. Lientz & Kathryn P. (1995) : Project Management for the 21st Century, Academic Press, California.

Earickson, R, and Harlin, J. (1994). : Geographic Measurement & Quantitative Analysis, Macmillan, New York.

W.E. Huxold & AG.Lerinsons Aronoft.S. (1989): Managing Geographic Information Projects.

PEDAGOGY:

Assignments pertaining to a relevant problem shall be given to students to apply the various aspects of research.

SEMESTER IV

PROJECT REPORT **Marks: 600**

(a) Mid – Term Appraisal of the Project Report	Marks: 100
(b) Evaluation of the Project Report	Marks: 400
(c) Presentation & Viva Voce (50+50)	Marks: 100

NOTE:

(i) Mid-term appraisal of the project report of each student shall be done by the board of the four examiners consisting of the supervisor, chairman, and the two senior faculty members of the department.

(ii) Mid-term appraisal of the project report shall have two-components: (i) presentation on project proposal including the research objectives, methodology and the chapter scheme, after the three weeks of assigning the project title, (ii) presentation of data analysis and tentative findings, after the six weeks of assigning the project title.

(iii) In case of project report, the Board of Control shall assign a candidate to a faculty member/s for supervision of his/her report on an approved topic. The student shall prepare three copies of his/her project report and submit in the concerned department. The report shall have to be submitted ten days before the commencement of the theory examination of the said semester. The project report shall incorporate the findings arrived at on the basis of the data/information collected from the field and processed, analyzed and mapped in the GIS lab.

For the evaluation of project report, three examiners, consisting of supervisor, chairperson of the department, and one more faculty member teaching the course, shall evaluate the field report, submitted by the student, independently and student shall be awarded the mean score of the three evaluations. The evaluation process shall have the two-components: (i) evaluation of report, (ii) presentation of data analysis and main findings before the board of examiners and viva voce.

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